

DIGITAL HEALTH

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Name of study: Master of Global Public Health

Topic: Total fertility rate (TFR): estimates for 1950-2021 and projections for 2022-2100

Why: Global TFR is projected to fall below the replacement level of 2.1 by 2030, which will directly alter the future age structure of populations. Therefore, policymakers and healthcare professionals must assess TFR trends to inform effective policy decisions.

What: An interactive dashboard that visualizes TFR trends using line charts and features five main functions: (1) a TFR definition popover, (2) country and region filters, (3) a selectable year range and a time-lapse feature, (4) hover-to-display data points, (5) chart and dataset download options.

How: The data were cleaned and processed using RStudio. The visualization dashboard was developed using Shiny, and the source code is shared via GitHub.

WHY?

Over more than 70 years, from 1950 to 2021, the global total fertility rate (TFR) declined by more than half, falling from 4.84 (4.63–5.06) to 2.23 (2.09–2.38). This downward trend is expected to continue, with global TFR potentially falling below the replacement level of 2.1 by 2030. In the long term, this may lead to negative population growth, as a TFR of 2.1 represents the rate at which a population replaces itself from one generation to the next.

Notably, some countries have experienced especially sharp declines. For example, in the Republic of Korea, TFR fell almost sevenfold, from 5.72 (5.37–6.08) in 1950 to just 0.82 (0.75–0.89) in 2021. Without effective government intervention, this decline is projected to continue through 2100.

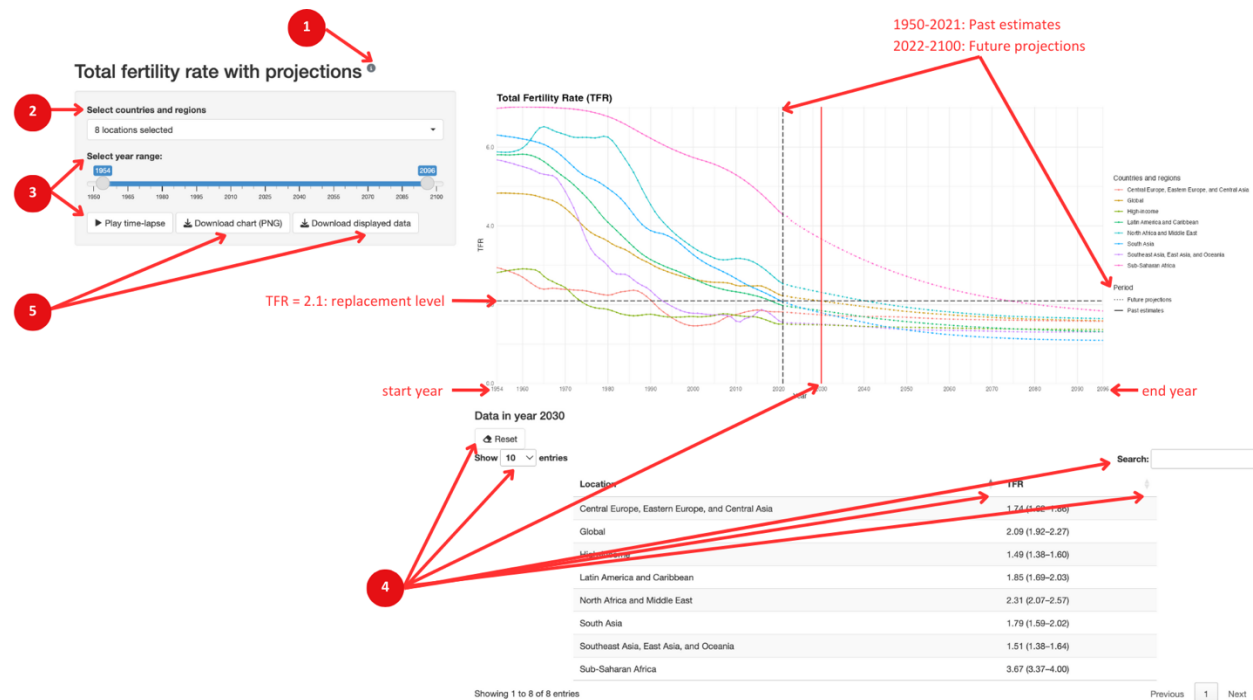
The TFR is a crucial demographic and public health indicator. It represents the average number of children a hypothetical cohort of women would have at the end of their reproductive period if they were subject during their whole lives to the fertility rates of a given period and if they were not subject to mortality.

Declining TFR will directly alter the future age structure of a country's population, with far-reaching implications for healthcare demand, labor supply, education systems, gender equality, family planning services, and social support structures.

Therefore, accurate assessment of current and projected TFR is essential for policymakers and healthcare professionals at both national and global levels to inform effective social, economic, and environmental policies.

This dashboard is based on estimates for the period 1950–2021, with projections for 2022–2100 provided by the Institute for Health Metrics and Evaluation (IHME). Its purpose is to visualize TFR trends using interactive line charts and features five main functions: (1) a TFR definition popover, (2) country and region filters, (3) a selectable year range and a time-lapse feature, (4) hover-to-display data points, (5) chart and dataset download options. These features enable policymakers and healthcare professionals to examine historical TFR trends, compare countries and regions, and anticipate TFR patterns over the coming decades.

WHAT?



1. Info circle icon

- First click: The TFR popover appears
- Second click: The TFR popover disappears

2. Select countries and regions

- Default: 8 priority regions are selected.
- Add or Remove Locations: Click to modify selections (maximum 50). If more than 50 locations are selected, names will be abbreviated to their first three letters.
- Select All: The dashboard remains active, but annotations will not display all 231 locations.
- Deselect All: Displays the warning: "No country or region selected. Try adding countries and regions to display data."

3. Select Year Range and Time-Lapse

- Select any start and end year.
- Use the "Play time-lapse" and then "Pause time-lapse" buttons to control the time-lapse.
- Automatic Reset: If the end year is set to 2100, clicking "Play time-lapse" will automatically reset the range to 1950–1951. Click "Play time-lapse" again to start the animation.

4. Hover over the chart to view data for a specific year

- Reset: Clear the selected year
- Entries per page: Show 10, 25, 50, or 100 entries
- Search: Find a specific location
- Sort: Organize by Location or TFR

5. Download chart and Download displayed data

- Click to download when necessary

HOW?

1. Data source

<https://ghdx.healthdata.org/record/ihme-data/global-burden-disease-study-2021-gbd-2021-fertility-1950-2100>

2. Data Cleaning

The source code is located in [cleaning_data.R](#)

The data cleaning process includes the following steps:

- Standardize column name syntax.
- Select relevant columns.
- Remove duplicate entries.
- Verify column classes (no need because they are already correct).
- Validate value coding (no need because they are already correct).
- Filter the dataset to "Reference" scenarios.

3. Shiny Dashboard

The source code is located in [shiny.R](#)

3.1. Data Preparation

- Group Locations: Categorize locations into priority regions (the 8 primary locations) and other locations.
- Year Range: Initialize the year range based on the minimum and maximum years present in the dataset.

3.2. User Interface (UI)

- Header: Include the dashboard title and a information icon with a TFR popover.
- Sidebar Layout:
 - Sidebar Panel: Includes a multi-select input for countries/regions, a slider for year range, and action buttons for Play Time-lapse, Download Chart, and Download Data.
 - Main Panel: Displays the reactive plot and the hover-data output.

3.3. Server

- State Management: Track and store the most recently hovered year.
- Data Transformation: Filter data based on selected year ranges and abbreviate location names if more than 50 are selected.
- Reactive Rendering: Generate the plot (including an "empty state" if no location is selected).
- Render the dynamic hover title and data table.
- Handle file download requests and time-lapse animations.

3.4. Run the Shiny application

4. Pushing Code to GitHub